



RAG-Based Semantic Search for Data

Unlocking a new dimension of data discovery through Retrieval-Augmented Generation.

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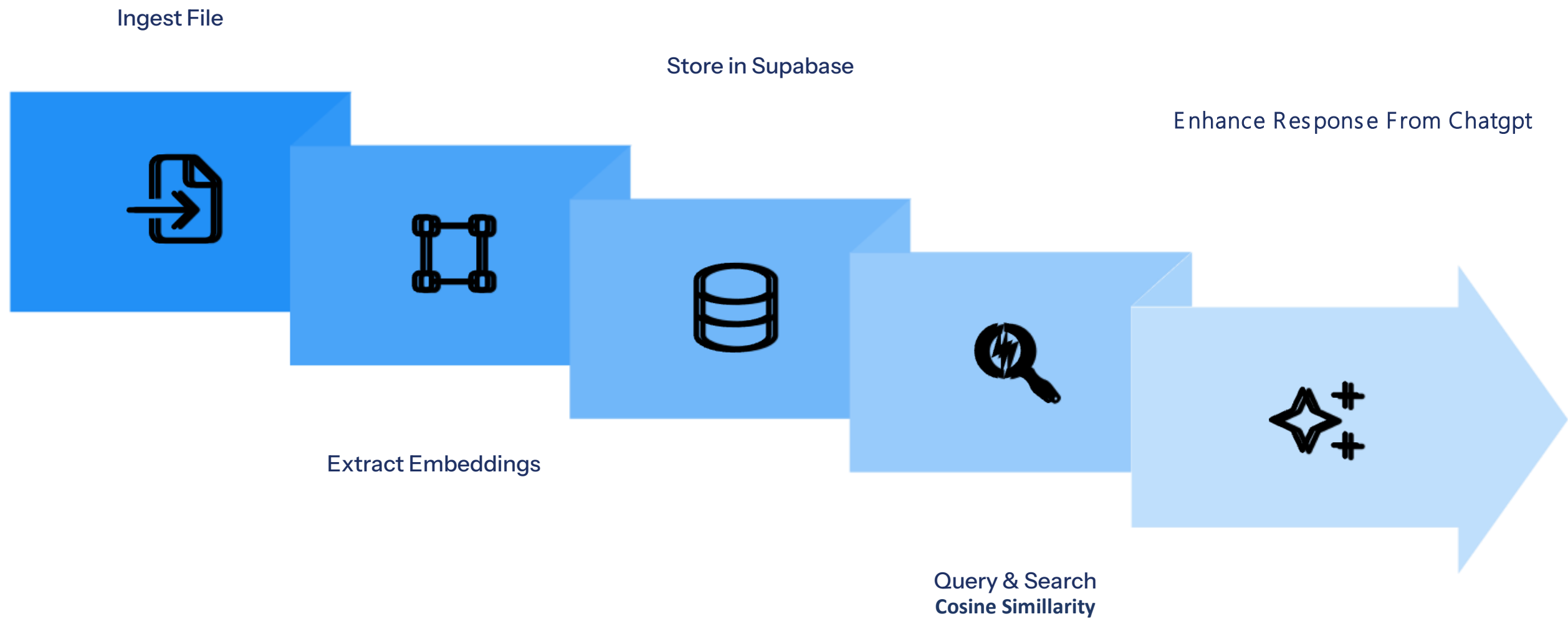
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System Design

This project leverages a sophisticated RAG pipeline to transform how we interact with movie data. Here's the blueprint:



Core Functionality

At its heart, our RAG system excels in precise data handling and intelligent response generation.

Data Preprocessing

Read `movies.txt`, chunk into smaller pieces. Convert each chunk into a **384-dimensional vector** using **gte-small** embedding model. Store vectors in **Supabase** for rapid retrieval.

Query Handling

User queries are transformed into vector embeddings. These are then compared with stored embeddings via **cosine similarity** for relevance.

Result Retrieval & Response Generation

Top N matching movie descriptions are fetched. These, along with the original query, are sent to **ChatGPT** to craft a creative and contextual reply.

Advantages & Limitations

✔ Merits ✔

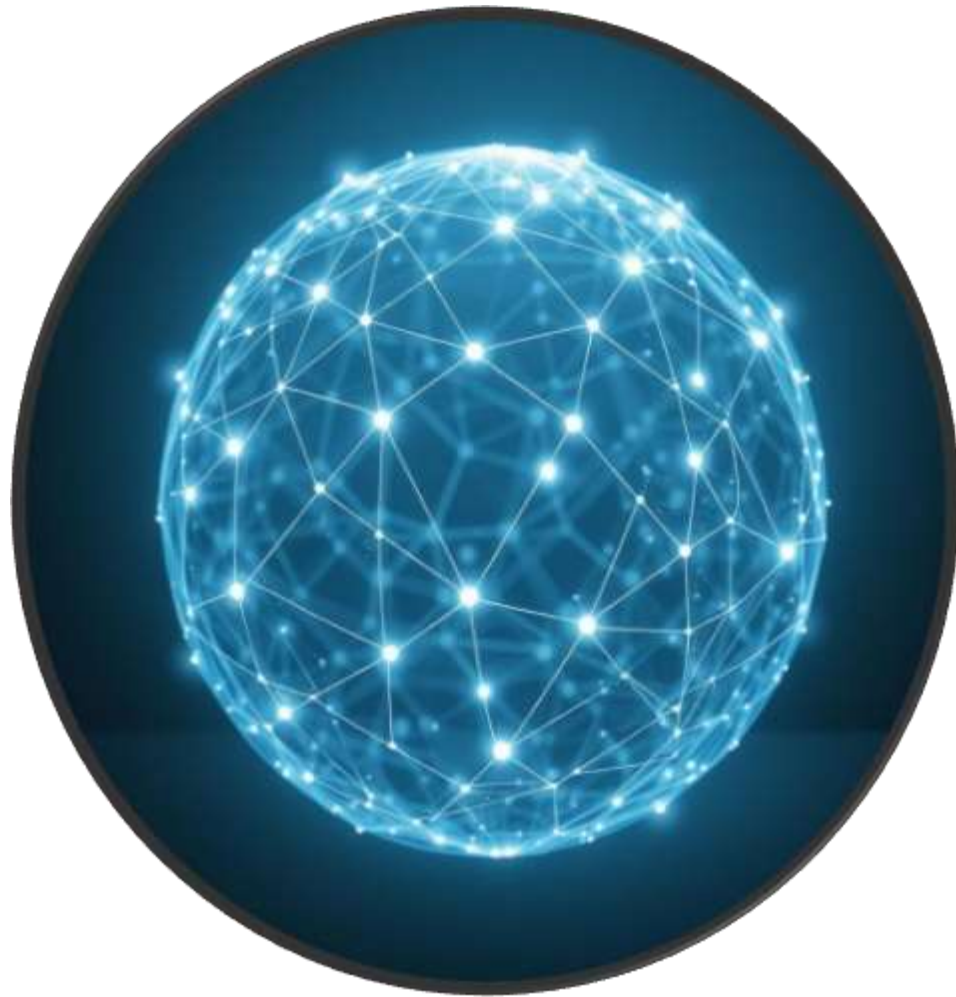
- High relevance from **semantic search**.
- **Creative & factual** outputs.
- Leverages **open-source embeddings** (cost-effective).
- **Scalable** with Supabase.

✘ Demerits ⚠

- **Dependent on embedding quality**.
- **Irrelevant results** possible with poor embeddings.
- **Requires internet** for ChatGPT API.
- Performance varies with **dataset size**.

Applications of Semantic Search

Semantic search extends far beyond movie data, revolutionizing how we interact with information across diverse industries. Its ability to understand context and meaning enables more intuitive and relevant results in various fields.



E-commerce

Enhancing product discovery with natural language queries.



Customer Support

Automating answers to complex user questions from knowledge bases.



Content Discovery

Providing relevant news, articles, and research based on user intent.



Healthcare

Retrieving precise medical information from vast datasets.



Education

Personalizing learning paths and resource recommendations.